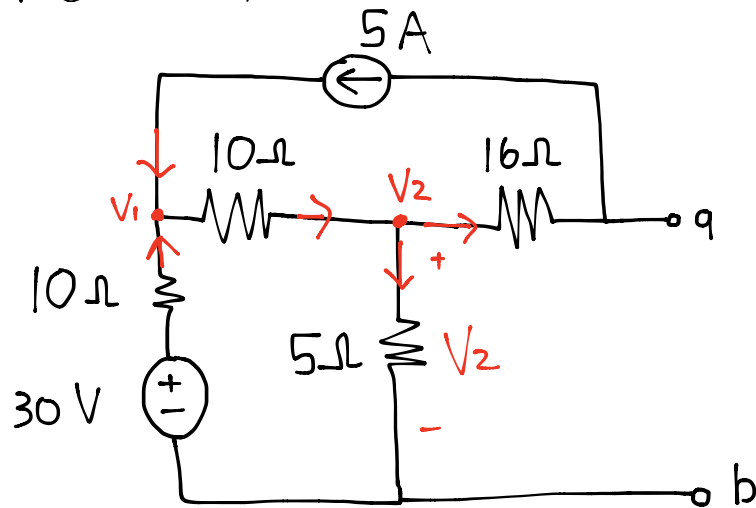


Tutorial 6

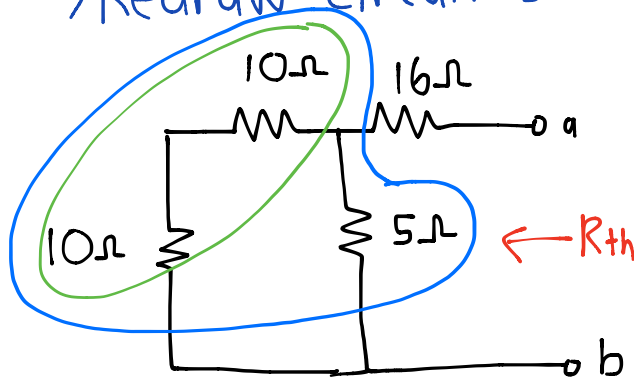
31 March 2016

4.39 Obtain the thevenin equivalent at terminals a-b of the circuit

* choose nodes
before starting
question



→ Redraw circuit switching off sources.



→ Serie = $10 + 10 = 20$

→ Parallel = $20 // 5 = \frac{20 \times 5}{20 + 5} = 4$

→ Final series

$R_{th} = 16 + 4 = 20\Omega$

→ Look at sources for V_{TH}

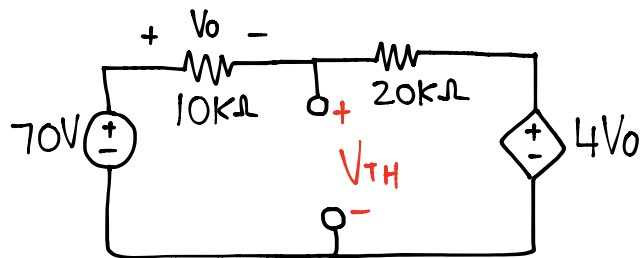
⇒ Node 1 : $\frac{30 - V_1}{10} + 5 = \frac{V_1 - V_2}{10} \rightarrow 80 = 2V_1 - V_2 \dots (1)$

⇒ Node 2 : $\frac{V_1 - V_2}{10} = 5 + \frac{V_2}{5} \rightarrow 100 = 2V_1 - 6V_2 \dots (2)$

* Subtracting (1) from (2) ⇒ $20 = -5V_2 \Rightarrow V_2 = -4V$

* KVL at V_{TH} : $-V_2 + 16 \times 5 + V_{TH} = 0 \Rightarrow \underline{V_{TH} = -84V}$

4.40 Find the thevenin equivalent at terminals a-b



* Apply KVL over the whole loop

$$-70 + (10k + 20k) \times I + 4V_o = 0$$

$$\Rightarrow V_o = 10k \times I$$

$$\therefore 70 = 30k \times I + 4 \times 10k \times I$$

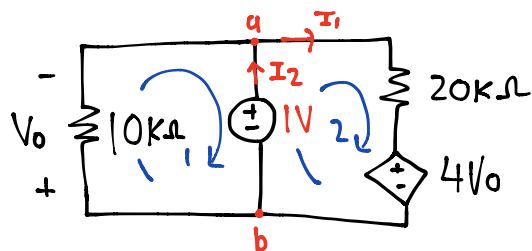
$$\Rightarrow 70 = 70k \times I \Rightarrow I = 1mA$$

* Apply KVL in inner loop to obtain V_{TH}

$$-70 + 10k \times I + V_{TH} = 0$$

$$V_{TH} = 70 - 10k \times 1m = \underline{60V}$$

* To find R_{TH} we must remove the source and apply a test source



* choose a 1V test source to make the maths easier

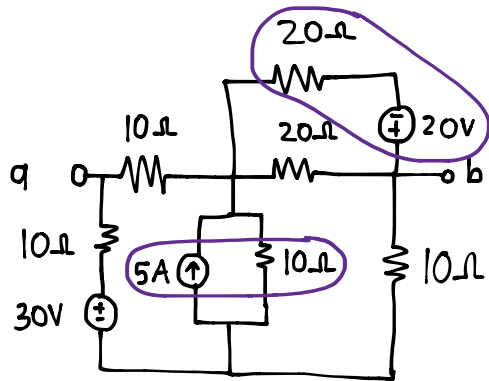
→ Notice that $V_o = -1V$

→ KVL in the 2nd loop $-1 + 20k \times I_1 + 4V_o = 0 \therefore I_1 = 0.25mA$

→ KCL at node a : $I_2 = I_1 + \frac{1V}{10k} = \underline{0.35mA}$

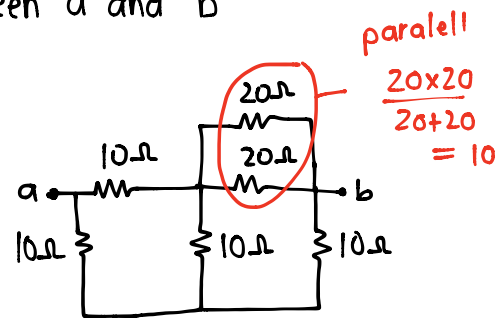
$$\therefore R_{TH} = \frac{1V \text{ (test source)}}{I_2} = \frac{1}{0.35 \times 10^{-3}} = \underline{2.857k\Omega}$$

4.42 Find the Thevenin equivalent between a and b



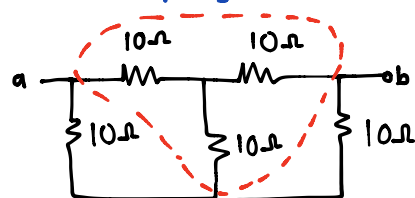
Switch off sources
 $\Rightarrow$

How to find R_{TH}

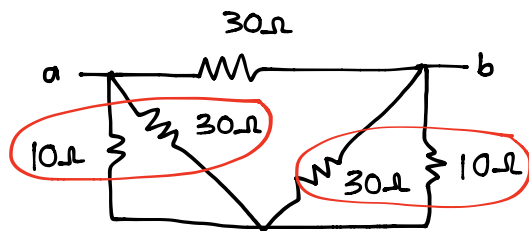


parallel
 $\frac{20 \times 20}{20 + 20} = 10$

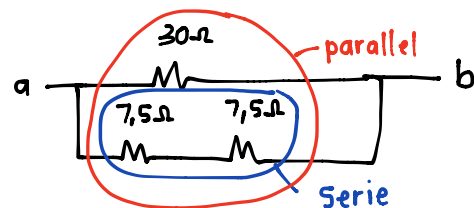
Simplify circuit



Y + Δ transform



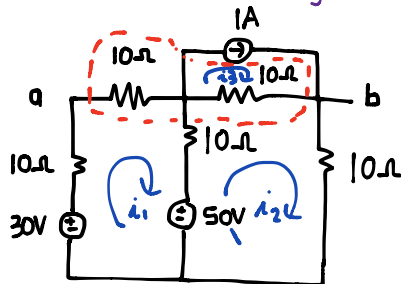
$\Rightarrow$ parallel $\frac{10 \times 30}{10 + 30} = 7,5$



$\therefore R_{TH} = 30 \parallel (7,5 + 7,5)$
 $= 10\Omega$

* To find V_{TH} use source Transformation

① Current to Voltage transform



* NB
 $i_3 = 1A$

* KVL in Loop 1:

$$-30 + 10i_1 + 10i_1 + 10(i_1 - i_2) + 50 = 0$$

$$30i_1 - 10i_2 = -20 \dots (1)$$

* KVL in Loop 2:

$$-50 + 10(i_2 - i_1) + 10(i_2 - i_3) + 10i_2 = 0$$

$$-50 - 10i_1 + 30i_2 - 10 = 0$$

$$-5 - i_1 + 3i_2 - 1 = 0$$

$$i_1 = 3i_2 - 6 \dots (2)$$

* Sub eq 2 into 1 $\therefore$ solve $i_2 = 2A$
 $i_1 = 0A$

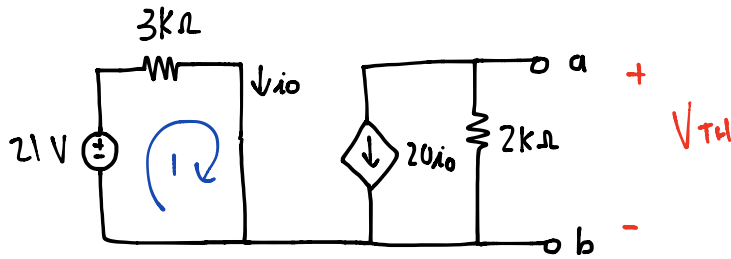
* NB: $V_{ab} = V_{TH}$ (Voltage across 2 10Ω resistors)

$$V_{th} = V_{10\Omega} + V_{10\Omega}$$

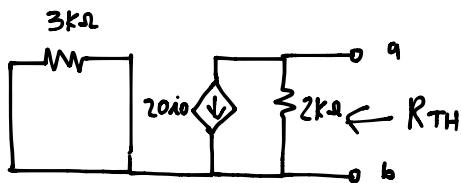
$$= 10i_1 + 10(i_2 - i_3)$$

$$= 10(0) + 10(2 - 1) = 10V$$

Q. 52 For the transistor model obtain the Thevenin equivalent at terminals a-b



* For the R_{TH} switch off sources



→ Due to no source in loop 1

$i_o = 0$ ∴ current source is also off

$$R_{TH} = 2k\Omega$$

* For V_{TH} look at the full circuit

→ KVL Loop 1 :

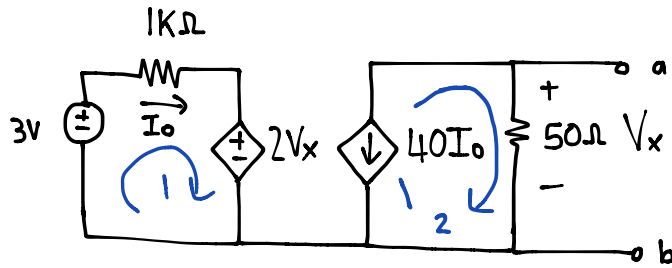
$$21 = 3k \times i_o \quad \therefore i_o = \underline{7mA}$$

$$V_{TH} = -20 \times i_o \times 2k$$

$$= -20 \times 7mA \times 2k$$

$$= \underline{-280V}$$

4.54 Find Thevenin equivalent between a-b



* $V_{TH} = V_x$ between a and b

* KVL Loop 1

$$-3 + 1k \times i_o + 2V_x = 0$$

* Loop 2 analysis

$$V_x = 50 \times (-40I_o)$$

combining equations

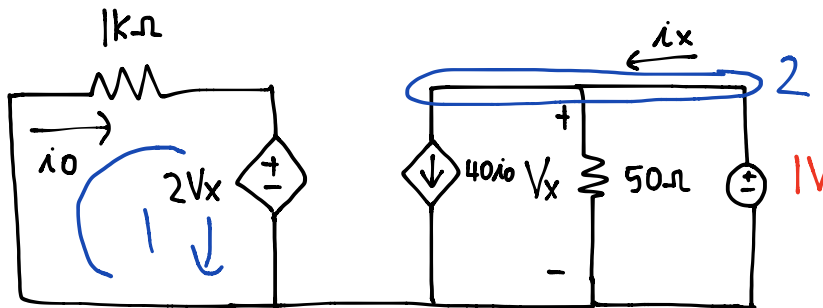
$$3 = 1000i_o + 2[-50 \times 40I_o]$$

$$\therefore I_o = -1mA$$

$$V_x = -2000(-1m) = 2$$

$$\therefore \underline{V_{TH} = 2V}$$

* To find R_{TH} switch off the source and implement test source (1V)



* Due to the test source in parallel with V_x

$$\underline{V_x = 1V}$$

$\Rightarrow$ Looking at loop 1

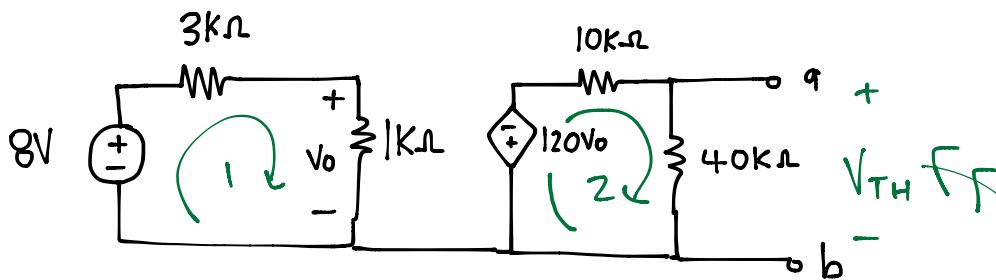
$$i_o = \frac{-2V_x}{1000} = -2mA$$

$\Rightarrow$ Looking at node 2 (KCL)

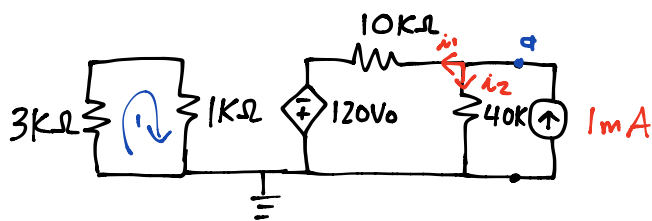
$$i_x = 40i_o + \frac{V_x}{50} = -80m + \frac{1}{50}$$

$$= -60mA \therefore R_{TH} = \frac{1}{i_x} = \frac{1}{-0.06} = \underline{-16.67\Omega}$$

4.71 For the circuit, what resistor connected across a-b terminals will absorb maximum power from circuit? What is the power?



* To find power we must find V_{TH} and R_{TH} at a and b
 $\Rightarrow$ Redraw circuit: (with test source at a-b terminal)



* For V_{TH} look at the sources *

* KCL at node a:

$$1 = \frac{V_a}{40} + \frac{V_a + 120V_o}{10} \Rightarrow 40 = 5V_a + 480V_o \dots (1)$$

* There is no Voltage source in Loop 1 $\therefore V_o = 0 \dots (2)$

* Sub (2) in (1) $\therefore \underline{V_a = 8V}$

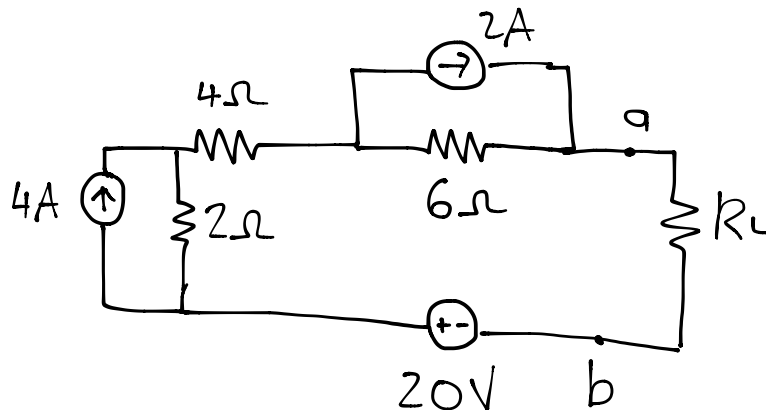
* $R_{TH} = V_a / 1mA = \underline{8k\Omega}$ $\therefore$ Resistance required

* Loop 1 $\therefore V_o = \frac{1}{4} \times 8 = 2V$ (Voltage division)

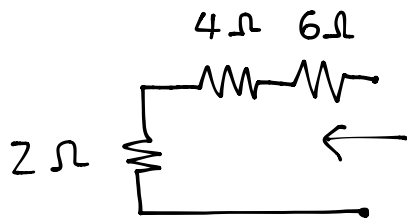
* Loop 2 $\therefore V_{TH} = \frac{40}{90} \times (-120V_o) = -192V$

* power = $V_{TH}^2 / 4R_{TH} = (-192)^2 / (4 \times 8 \times 10^3) = 1.152 \text{ watts}$

- 4.72
- Obtain Thevenin equivalent at a-b
 - Calculate current in $R_L = 13\Omega$
 - Find R_L for maximum power deliverable to R_L
 - Determine the maximum power

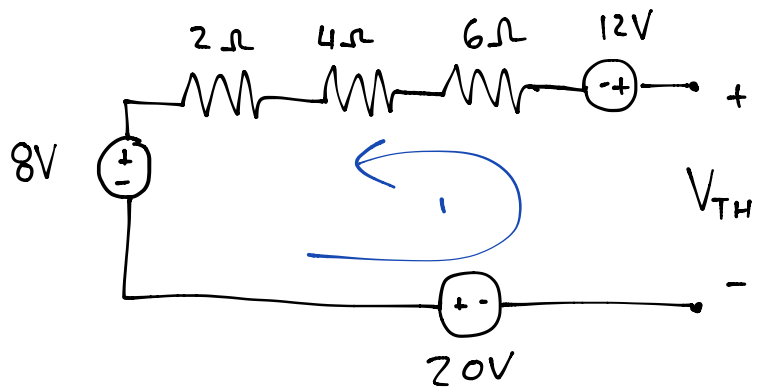


* a) Redraw circuit



$$R_{TH} = 2 + 4 + 6 = \underline{12\Omega}$$

Transform sources



* Add Voltages in loop

$$-V_{TH} + 12 + 8 + 20 = 0$$

$$\therefore \underline{V_{TH} = 40V}$$

*b) Current

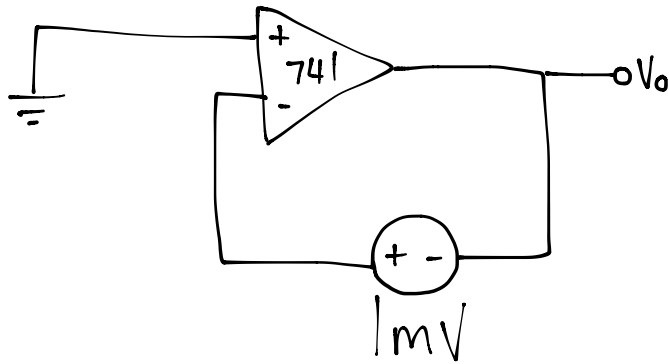
$$i = \frac{V_{TH}}{(R_{TH} + R)} = \frac{40}{(12 + 13)} = \underline{1.6A}$$

* c) For maximum power transfer

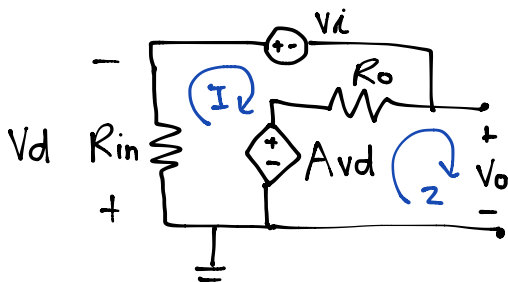
$$R_L = R_{TH} = \underline{12\Omega}$$

$$* d) \text{ power} = \frac{V_{TH}^2}{4R_{TH}} = \frac{40^2}{(4 \times 12)} = \underline{33.33 \text{ Watt}}$$

5.6 Using the same parameters as 741 opamp in Ex.1
Find v_o of the circuit below:



→ To obtain v_o we must look at the equivalent circuit:



*KVL in Loop 1:

$$(R_i + R_o)I + V_i + A_{vd} = 0 \dots (1)$$

*NB $V_d = R_i \times I$

$$\Rightarrow \text{Sub in (1)} \quad V_i + (R_i + R_o + R_i A)I = 0$$

→ Get I alone

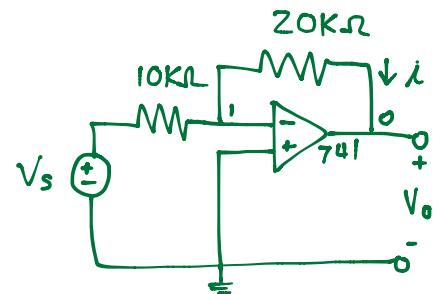
$$I = \frac{-V_i}{R_o + (1+A)R_i} \dots (2)$$

Ex.1

→ open-loop voltage gain = 2×10^5

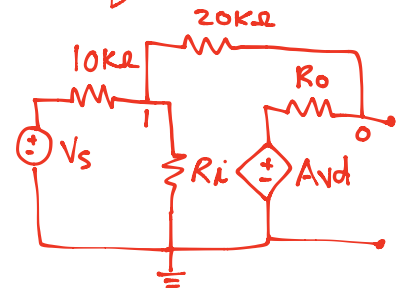
→ Input resistance = $2\text{M}\Omega$

→ output resistance = 50Ω



? i when $V_s = 2\text{V}$

⇒ Equivalent circuit



→ KVL in Loop 2:

$$-A v_d - R_o I + V_o = 0$$

$$\begin{aligned} V_o &= A v_d + R_o I \\ &= (R_o + R_i A) I \dots (3) \end{aligned}$$

→ Sub (2) into (3)

$$\begin{aligned} V_o &= (R_o + R_i A) \left[\frac{-V_i}{R_o + (1+A)R_i} \right] \\ &= -V_i \left(\frac{R_o + R_i A}{R_o + (1+A)R_i} \right) V_i \end{aligned}$$

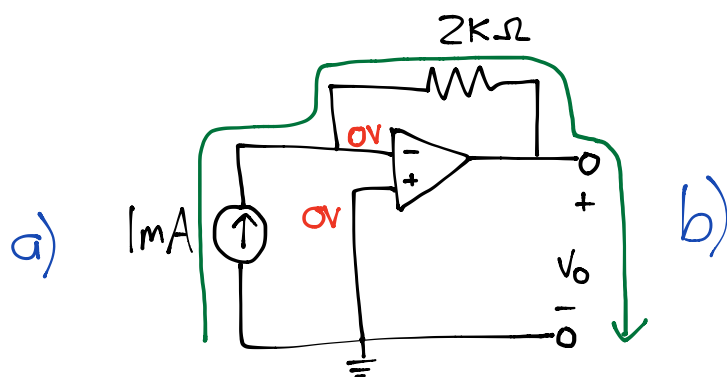
⇒ Sub values $R_o = 50$ $R_i = 2M$ $A = 2 \times 10^5$ $V_i = 1 \times 10^{-3}$

$$V_o = - \frac{(50 + 2 \times 10^6 \times 2 \times 10^5) \times 10^{-3}}{50 + (1 + 2 \times 10^5) \times 2 \times 10^6}$$

$$= \frac{-200,000 \times 2 \times 10^6}{200,001 \times 2 \times 10^6}$$

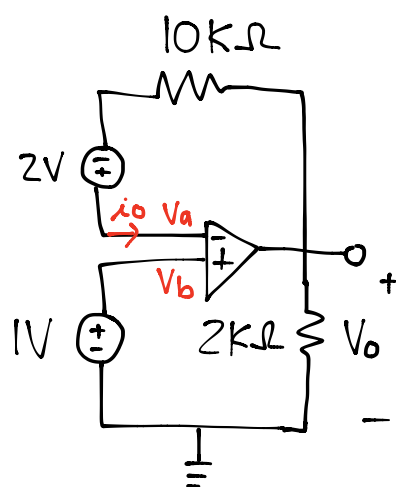
$$= \underline{\underline{-0.999995 \text{ mV}}}$$

5.8 Obtain V_o for each op amp



→ Due to inverting and noninverting terminals $V_a = V_b = 0V$
 → current flow

$$1mA = \frac{0 - V_o}{2k} \therefore V_o = -2V$$

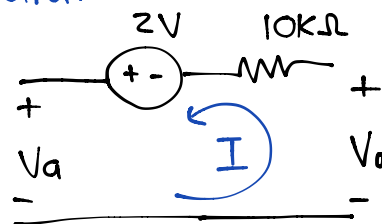


* $V_a = V_b = 1V$
 → No current at V_a

$$i_o = 0$$

* There is no current through $10k\Omega$

⇒ Redraw circuit



→ Looking at Loop

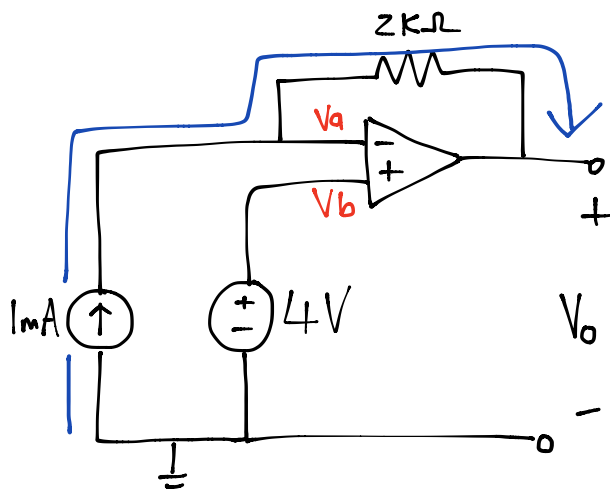
$$-V_a + 2 + V_o = 0$$

$$V_o = V_a - 2$$

$$= 1 - 2$$

$$= -1V$$

5.9 Determine V_o for op amps

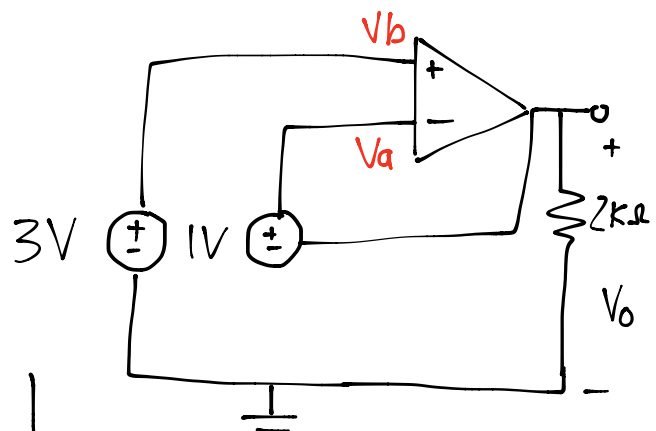


$$*V_a = V_b = 4V$$

$\Rightarrow$ Current flow at inverting terminal:

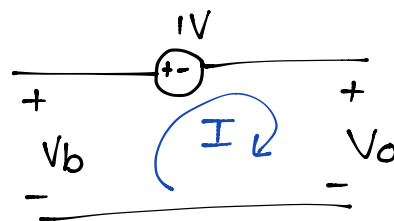
$$1mA = \frac{4 - V_o}{2k}$$

$$\therefore V_o = \underline{2V}$$



$$V_a = V_b = 3V$$

$\rightarrow$ Redraw



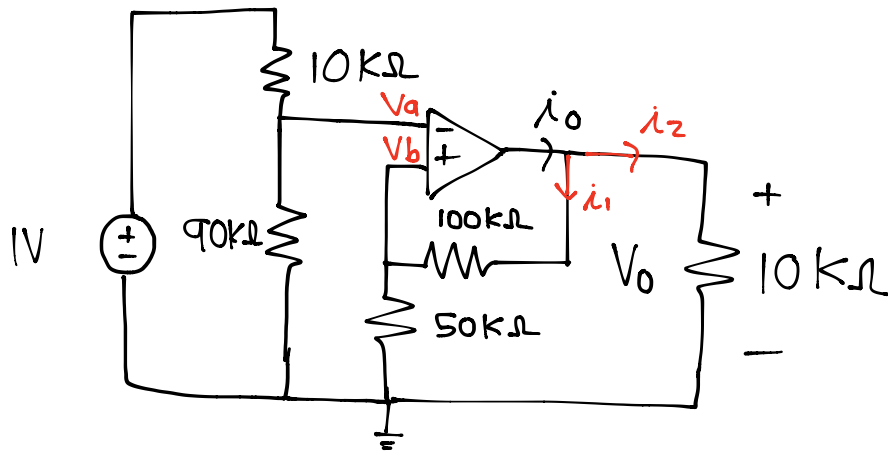
$\rightarrow$ KVL in Loop

$$-V_b + 1 + V_o = 0$$

$$V_o = 3 - 1$$

$$= \underline{2V}$$

5.13 Find V_o and i_o in circuit



* Use voltage division

$$V_a = \frac{90}{100}(1) = \underline{0.9V}$$

$$V_b = \frac{50}{100} \times V_o = \underline{\frac{V_o}{2}}$$

* NB
 $V_a = V_b$

$$\frac{V_o}{2} = 0.9$$

$$\therefore V_o = \underline{2.7V}$$

* Current at node:

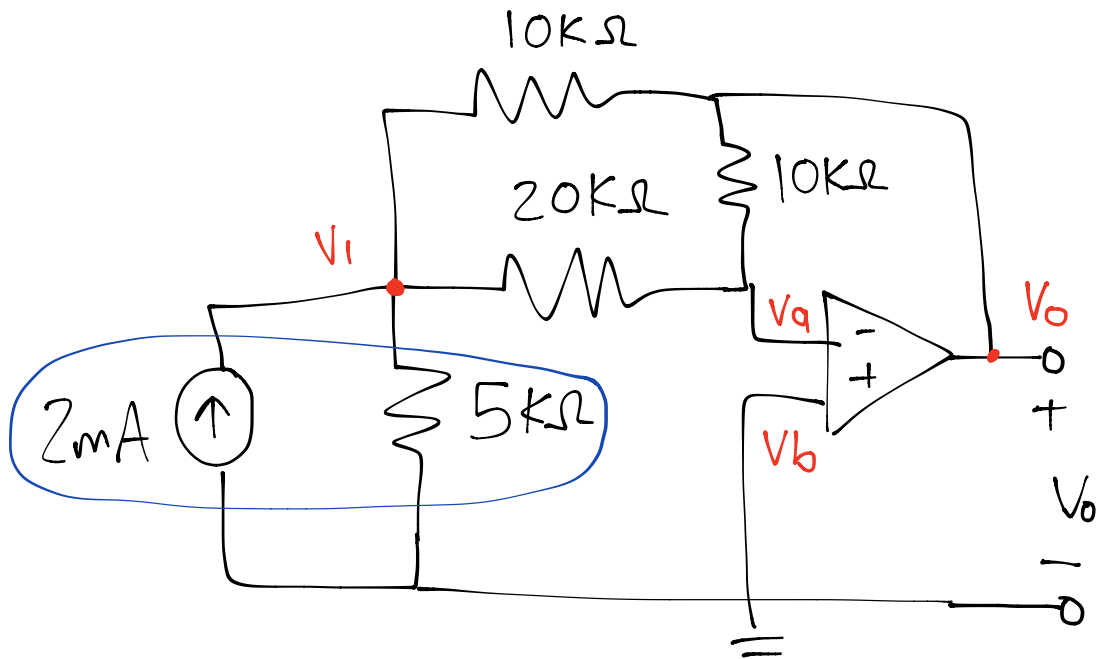
$$i_o = i_1 + i_2$$

$$= \frac{V_o}{150k} + \frac{V_o}{10k}$$

$$= 0.27m + 0.018m$$

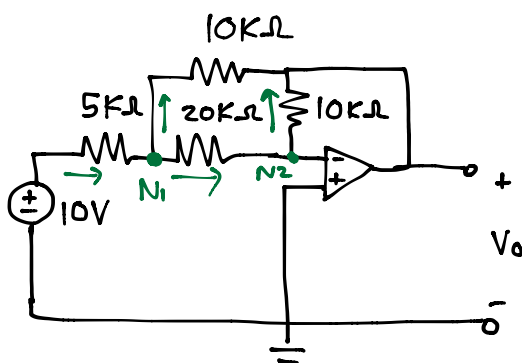
$$= 288 \mu A$$

5.14 Determine the output voltage V_o



① Transform circuit for analysis

→ Redraw circuit:



• Nodal analysis

$$N_1: \frac{10 - V_1}{5} = \frac{V_1 - V_a}{20} + \frac{V_1 - V_o}{10}$$

NB $V_b = V_a = 0V$

$$\therefore 40 = 7V_1 - 2V_o \dots (1)$$

$$N_2: \frac{V_1 - V_a}{20} = \frac{V_a - V_o}{10}$$

$$\therefore V_1 = -2V_o \dots (2)$$

*Sub (1) and (2)

$$40 = 7(-2V_o) - 2V_o$$

$$\therefore V_o = \underline{-2.5V}$$